Inner product space V

over $F = \mathbb{R}$ or $\mathbb{C}$ has inner prod

$V \times V \xrightarrow{\langle \cdot, \cdot \rangle} F$ satisfying

- $\langle v, v \rangle$ real ≥ 0 with $\langle v, v \rangle = 0 \Leftrightarrow v = 0$
- $\langle v, w \rangle = \overline{\langle w, v \rangle}$ → complex conjugate
- $\langle cv_1 + v_2, w \rangle = \overline{c}\langle v_1, w \rangle + \langle v_2, w \rangle$
- $\langle v, cw_1 + w_2 \rangle = c\langle v, w_1 \rangle + \langle v, w_2 \rangle$

opposite to LADR

Examples · $V = F^n$ $\langle v, w \rangle := \overline{v}^T w$ check!

· Any subspace of an inner product space

· perpendicularity $\langle v, w \rangle = 0 \Leftrightarrow \langle w, v \rangle = 0$

For $S \subset V$, let $S^\perp = \{v \in V \mid v \perp s \ \forall s \in S\}$ is a subspace

· $\|v\| = \sqrt{\langle v, v \rangle}$ real and positive unless $v = 0$

For subspace W, we have $V = W \oplus W^\perp$ ($\text{so } (W^\perp)^\perp = W$)

· $W \cap W^\perp = 0$ by positive definiteness

· $\dim(W^\perp) \geq n - \dim W$ as $W^\perp$ = sol^n space of $\dim W$ homogeneous linear eq's after fixing some basis (Ex)

· Any set of mutually $\perp$ (nonzero) vectors is linearly indep

· Pythagoras: $v \perp w \Rightarrow \|v+w\|^2 = \|v\|^2 + \|w\|^2$

O.N. basis $\{e_1, e_2, \cdots, e_n\}$ means

$\langle e_i, e_j \rangle = \delta_{ij} \Rightarrow V = \sum \langle e_i, v \rangle e_i = \sum \overline{\langle v, e_i \rangle} e_i$

Any O.N. list can be completed to an O.N. basis using $V = W \oplus W^{\perp}$, or explicitly using Gram-Schmidt

Formulas. ① projection along a nonzero v

$$P_v(u) = \frac{\langle v, u \rangle}{\|v\|^2} v$$

$$u = P_v(u) + \underbrace{u - P_v(u)}_{\perp \text{ to } v}$$

Consequences of ① Check these!

② Cauchy Schwarz $\Rightarrow$ ③ Δ inequality

$|\langle v, u \rangle| \leq \|u\| \|v\|$

with equality iff

$u \in Fv$ ← (Assume $v \neq 0$ WLOG) → $u \in \mathbb{R}_{\geq 0} v$.

Proof using O.N basis?

④ Gram-Schmidt: stays same!

$v_1, \cdots, v_n$ any basis $\rightsquigarrow$ O.N. basis $e_1, \cdots, e_n$

$$e_1 = \frac{v_1}{\|v_1\|}, \cdots, e_{k+1} = \frac{v_{k+1} - \sum_{i=1}^{k} \langle e_i, v_{k+1} \rangle e_i}{\| \text{numerator} \|}$$

Check: $\langle e_j, e_{k+1} \rangle \stackrel{?}{=} 0$

Interesting operators on inner product spaces

① T s.t. $\langle Tv, Tw\rangle = \langle v, w\rangle$ name ($F=\mathbb{R}$: orthogonal) ($F=\mathbb{C}$: unitary)

a) Let v be an evector, say $Tv=\lambda v$ Then $\langle v,v\rangle = \langle Tv, Tv\rangle = \langle \lambda v, \lambda v\rangle = \bar{\lambda}\lambda\langle v,v\rangle \Rightarrow |\lambda| = 1$

b) Let W be a T-invariant subspace. Then $W^\perp$ is also T-invariant! (Ex. use that T is invertible)

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② T s.t. $\langle Tv, w\rangle = \langle v, Tw\rangle$ name: (symmetric $F=\mathbb{R}$ Hermitian $F=\mathbb{C}$)

a) Let v be an evector, say $Tv=\lambda v$ $\bar{\lambda}\langle v,v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \lambda\langle v,v\rangle \Rightarrow \lambda \in \mathbb{R}$

b) Same claim as above!

Pf let $u \in W^\perp$ ⇓ $0 = \langle u, W\rangle$

want $Tu \in W^\perp$ ie want $\langle Tu, w\rangle = 0 \ \forall w \in W$

but $\langle Tu, w\rangle = \langle u, Tw\rangle$ and $Tw \in W$ $= 0$

Rmk ① Fix an O.N. basis $e_1, \cdots, e_n$ $[v]$ = coord of v wrt this basis

$$\boxed{\langle v, w \rangle = \overline{[v]}^T [w]}$$

② Let T be any op on V

A = matrix of T wrt ON basis

So $[Tv] = A[v]$

$$\begin{cases} \langle Tv, Tw \rangle = \langle v, w \rangle \iff \bar{A}^T A = I & \text{(Unitary if } F = \mathbb{C} \text{, orthogonal if } F = \mathbb{R}\text{)} \\ \langle Tv, w \rangle = \langle v, Tw \rangle \iff \bar{A}^T = A & \text{(Hermitian matrix, Symmetric if } F = \mathbb{R}\text{)} \end{cases}$$

(ex.)

Notation for complex matrices $\boxed{M^* = \overline{M}^T}$

Spectral thms: Let T be a "good" operator on a finite dimensional inner product space V

V has an orthonormal basis of eigenvectors of T with "good" eigenvalues

Today $F = \mathbb{C}$. So T has an eigenvector. Then ⋯

Good operator = unitary / Hermitian

good evalue = $|\lambda| = 1$ $\quad \lambda \in \mathbb{R}$

Pf of spectral thms: use a, b and induct on dimension!

(Ex.) What happens when $F = \mathbb{R}$?

(Ex) Write matrix versions of all spectral thms.

a) Prove spectral thm for symmetric operators (hint: go via a matrix)

b) Prove appropriate thm for orthogonal operators: get O.N. basis wrt which matrix is "as diagonal as possible" (meaning 1×1 or 2×2 blocks)

(Ex.) Are converses of the four thms true?

(Ex.) A complex matrix M is called <u>normal</u> if $M^*M = MM^*$. Spectral thm for these??
Can you define normal <u>operator</u> on an inner product space? (We will see the answers!)

<u>Note</u>: Unitary matrix A has $A^* = A^{-1}$ and Hermitian A has $A^* = A$, so they are normal!